

Implementing and Evaluating KADABRA and BRAVA-GNN in Julia

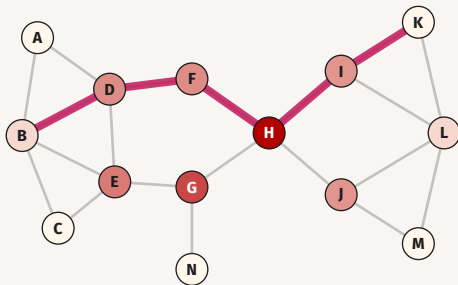
Lab Computational Analytics — Open-Source Graph Software Engineering

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Laura Bülte

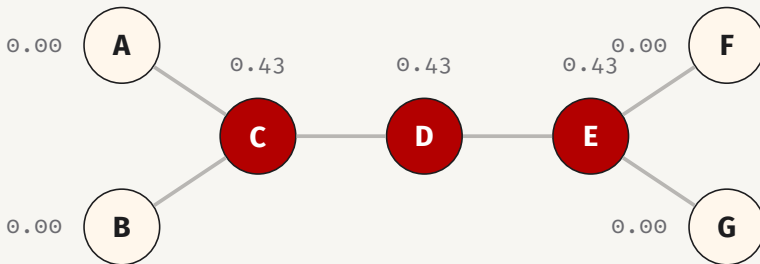
September 16, 2026



Betweenness Centrality

$$bc(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

Betweenness Centrality: Example



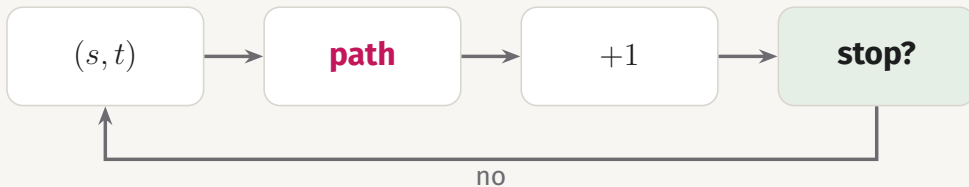
Exact Computation Does Not Scale

$$O(nm)$$

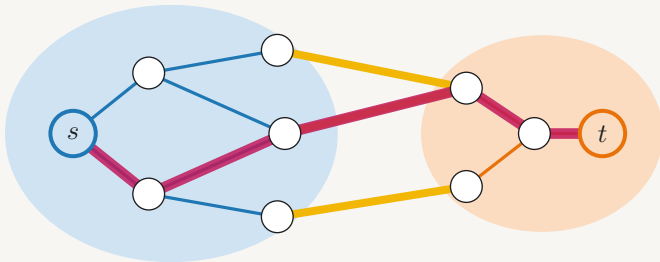
169 min

exact betweenness, email-EuAll

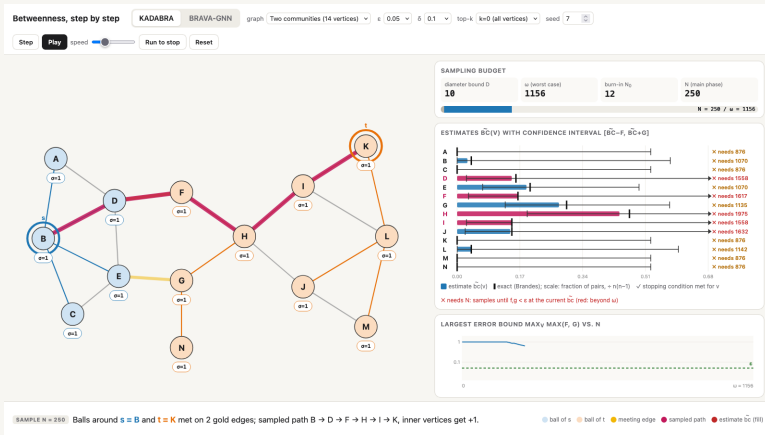
KADABRA: Sample Shortest Paths



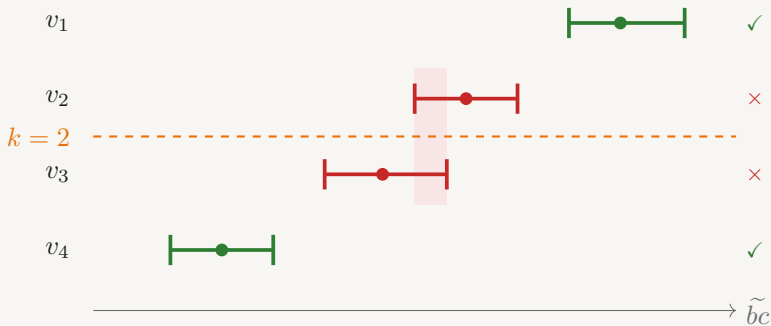
Sampling a Path: Balanced Bidirectional BFS



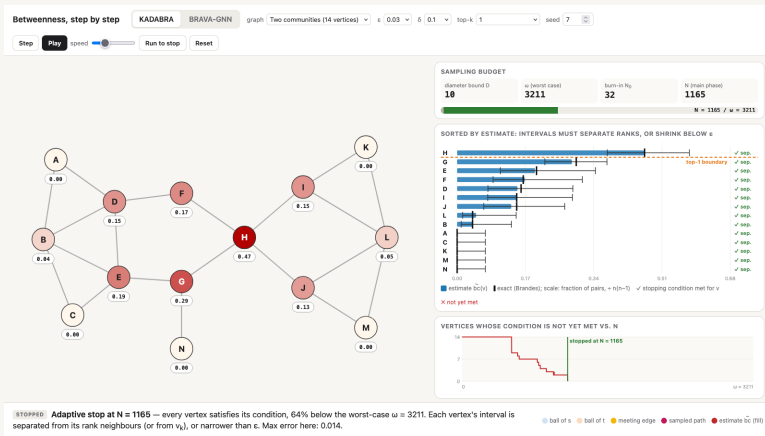
Live Demo: Sampling Paths


[Open live demo](#)

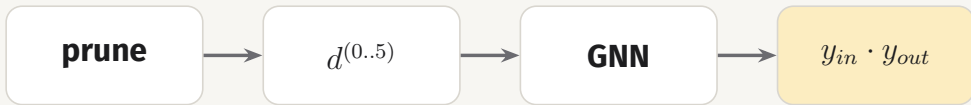
When to Stop: Top- k



Live Demo: Top- k Stops Early

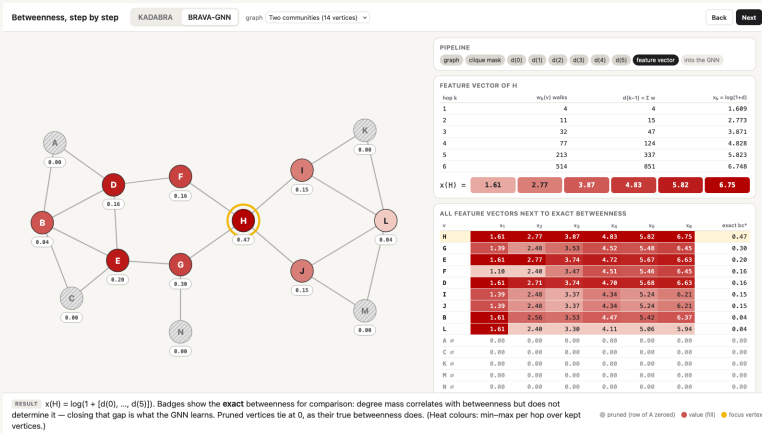

[Open live demo](#)

BRAVA-GNN: Learn the Ranking



≈ 1300 parameters

Live Demo: One Feature Vector


[Open live demo](#)

What We Did



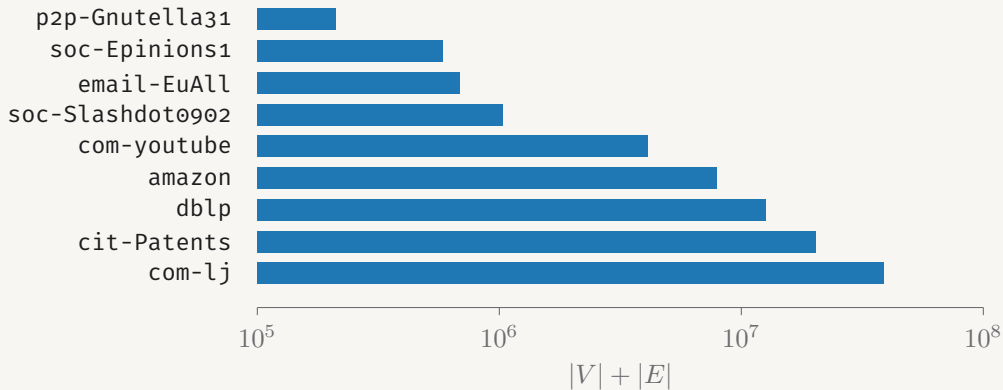
Research Questions

RQ1 Julia vs. C++

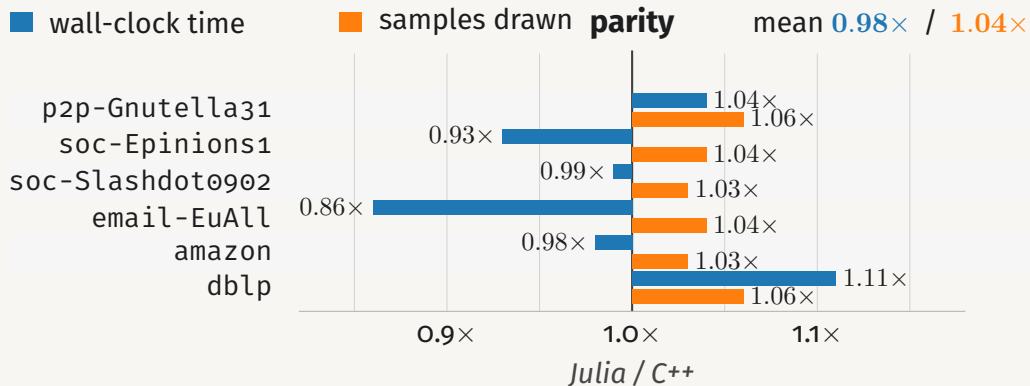
RQ2 BRAVA-GNN vs. KADABRA

RQ3 Does top- k save work?

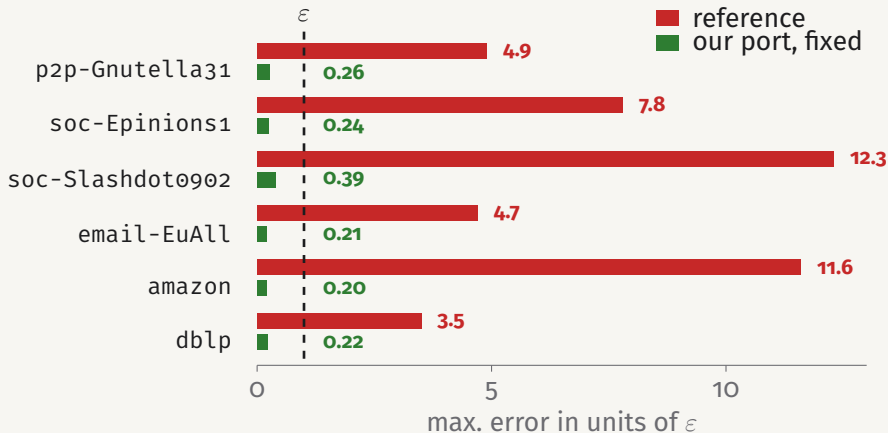
Setup



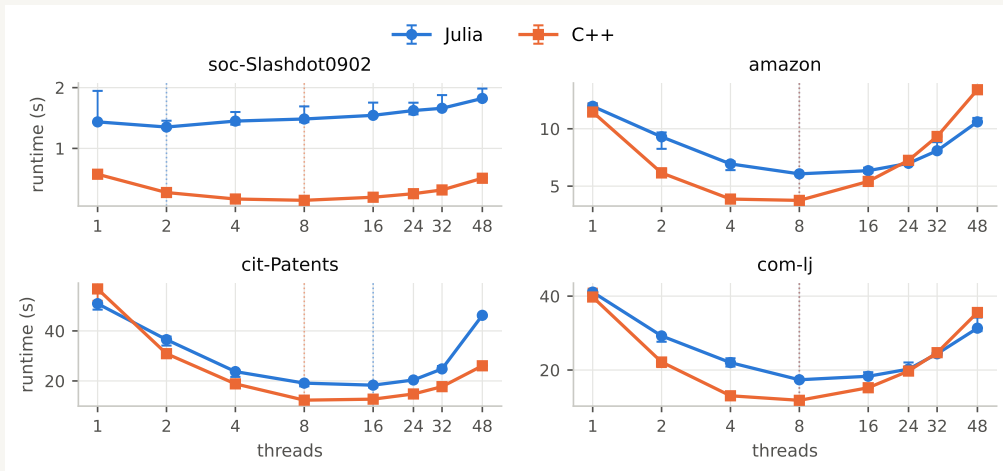
Same Samples, Same Time



A Bug We Copied: Scores Too Low



Thread Scaling



Answer to RQ1

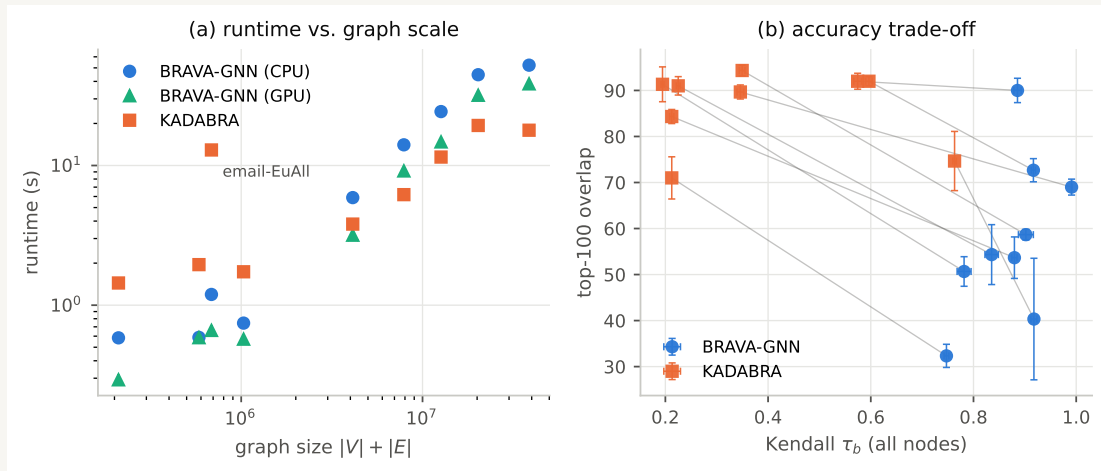
Julia $\approx$ C++

1.04× samples · 0.98× time

At $\varepsilon = 0.01$: Opposite Strengths

	BRAVA-GNN	KADABRA	
Kendall τ_b	0.87	0.39	9 of 9
top-100 overlap	58	87	9 of 9
runtime	0.6–52 s	1.4–19 s	

mean over 9 graphs

At $\varepsilon = 0.01$, Graph by Graph

At $\varepsilon = 0.0001$: KADABRA Wins Accuracy

	BRAVA-GNN	KADABRA	
Kendall τ_b	0.87	0.95	8 of 9
top-100 overlap	58	99.7	9 of 9
runtime	0.6–52 s	48 s–10.8 h	40–1365×

mean over 9 graphs · BRAVA-GNN unchanged

Answer to RQ2

BRAVA-GNN

fast

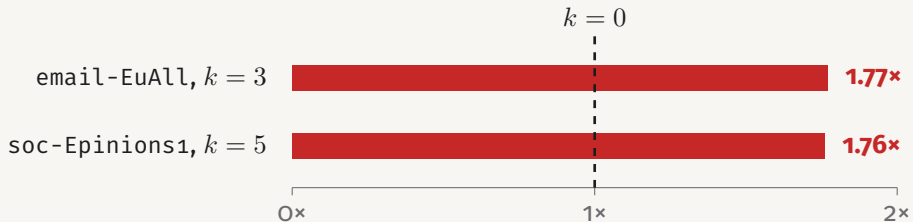
$0.6\text{--}52\text{ s} \cdot \tau_b$ 0.87

KADABRA

accurate

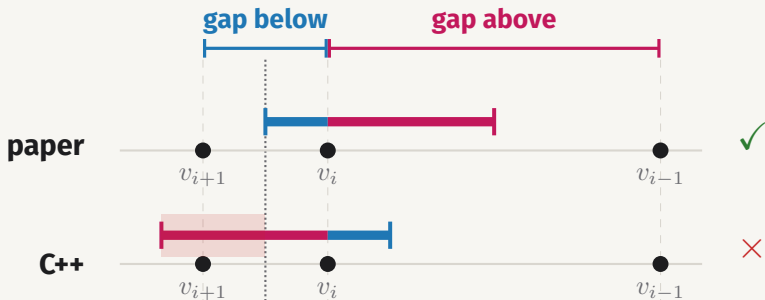
τ_b 0.95 · top-100 99.7

Asking for Less Costs More

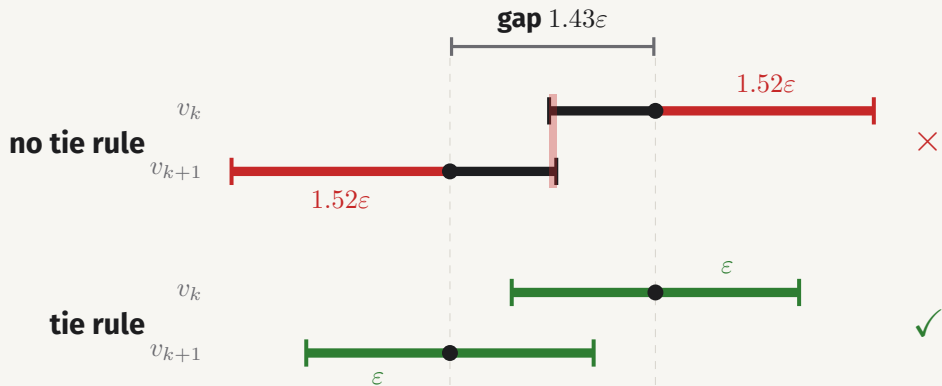


6 of 16 top- k configurations cost more than $k=0$

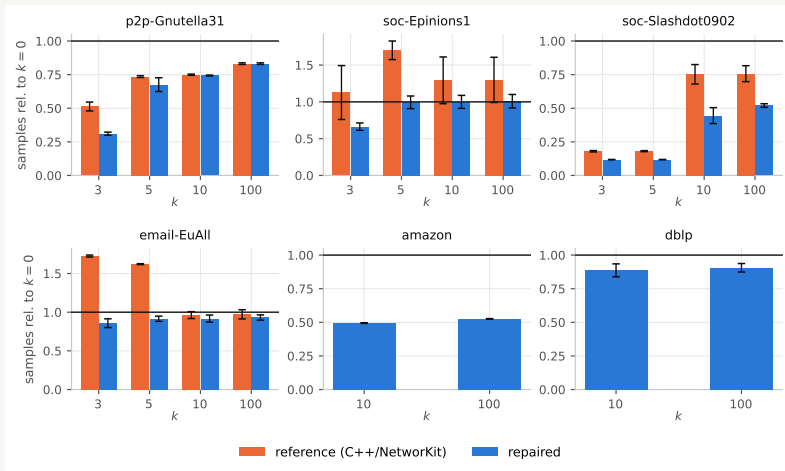
The Bug: Budgets Sized by the Wrong Gap



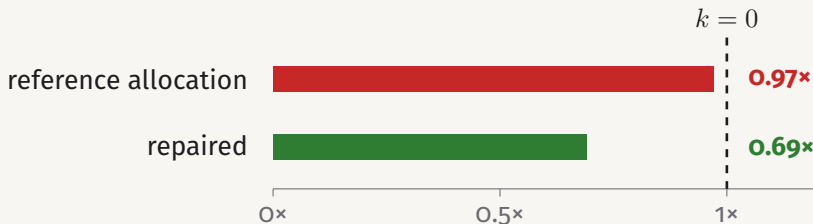
...and a Deadlock at the Top- k Boundary



Repaired: Top- k Never Costs More



Answer to RQ3



same top- k answers · fixed in `Graphs.jl` PR #518

Conclusion

RQ1 Julia $\approx$ C++

RQ2 speed or accuracy

RQ3 top- k saves work, once fixed

Limitations and Open Question

Who wins at a **fixed time budget**?

References

Algorithms

- [1] U. Brandes. A faster algorithm for betweenness centrality. *J. Math. Sociology* 25(2), 2001.
- [2] L. C. Freeman. A set of measures of centrality based on betweenness. *Sociometry*, 1977.
- [3] I. Pohl. Bi-directional search. *Machine Intelligence* 6, 1971.
- [4] M. Borassi, E. Natale. KADABRA is an ADaptive Algorithm for Betweenness via Random Approximation. *ACM JEA* 24, 2019.
- [5] J. Dachille, A. Rossi, S. K. Maurya, F. Mallmann-Trenn, X. Liu, F. Giroire, T. Murata, E. Natale. Degree-Mass Message Passing for Betweenness Ranking in Directed and Undirected Networks. *CIKM '26*, 2026.

Data and software

- [6] J. Leskovec, A. Krevl. SNAP Datasets. snap.stanford.edu/data, 2014.
- [7] C. L. Staudt, A. Sazonovs, H. Meyerhenke. NetworkKit: A tool suite for large-scale complex network analysis. *Network Science* 4(4), 2016.
- [8] Graphs.jl, github.com/JuliaGraphs/Graphs.jl; KADABRA pull request #518.

Thank you

questions?